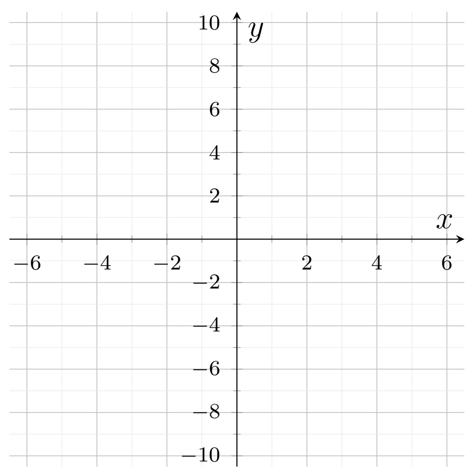
Graphing Quadratics from Vertex or Standard Form

Reading a parabola from vertex form, from standard form, and from its intercepts

Directions. Every grid runs from -6 to 6 across and -10 to 10 up and down. For each function, find the features asked for, then sketch the parabola on its grid using the vertex, the axis of symmetry, and at least two more points.

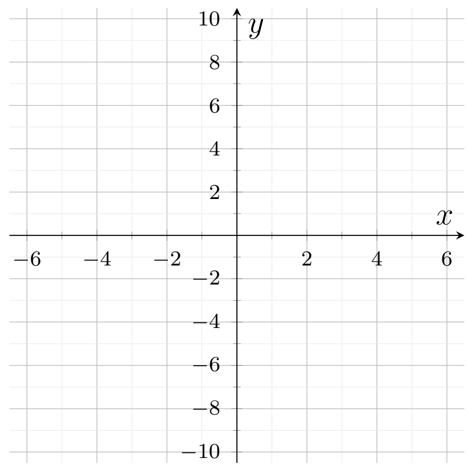
- **Vertex form** $y = a(x-h)^2 + k$: the vertex is (h, k). Read the sign carefully — $(x+3)^2$ means $h = -3$.
- **Standard form** $y = ax^2 + bx + c$: the axis of symmetry is $x = -\frac{b}{2a}$; substitute it back to get the vertex y-value. The y-intercept is c.
- $a > 0$ opens up, $a < 0$ opens down; $|a| > 1$ is narrower and $|a| < 1$ is wider than $y = x^2$.

1. $y = (x-2)^2 - 1$. Give the vertex and say which way the parabola opens, then sketch it.



Answer: ____________

2. $y = -(x+3)^2 + 4$. Give the vertex, say which way the parabola opens and why, then sketch it.



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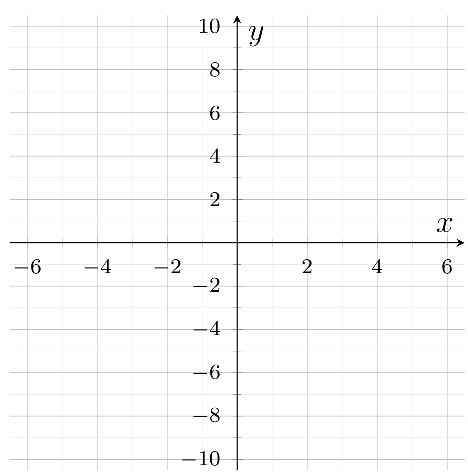
3. $y = x^2 - 6x + 5$. Use $x = -\frac{b}{2a}$ to find the axis of symmetry, then give the vertex and the y-intercept, and sketch.



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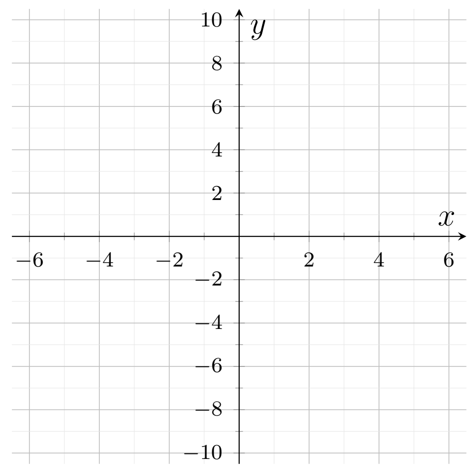
4. $y = x^2 - 4x - 5$. Factor to find the x-intercepts, then sketch the parabola through them.



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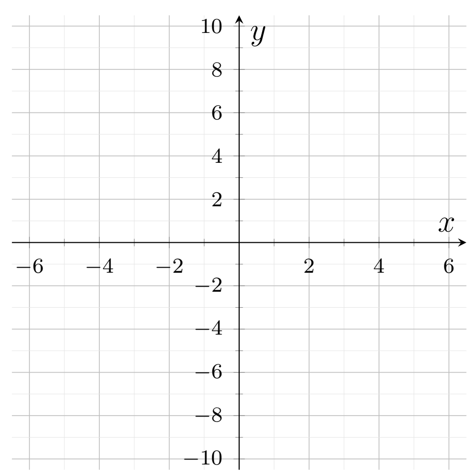
5. $y = 2(x-1)^2 - 8$. Give the vertex, then solve $2(x-1)^2 - 8 = 0$ for the two x-intercepts and sketch.



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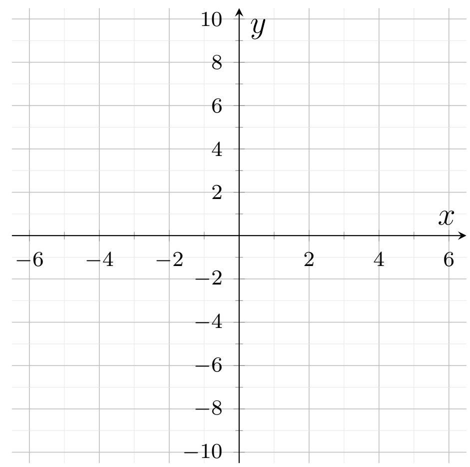
6. $y = -x^2 + 4x + 5.$ Find the vertex and the y-intercept. Is the vertex a maximum or a minimum? Sketch the parabola.



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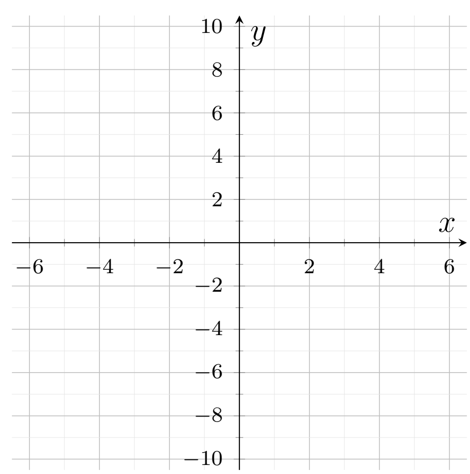
7. $y = x^2 - 9.$ Give the x-intercepts. Where does the axis of symmetry sit, and why? Sketch.



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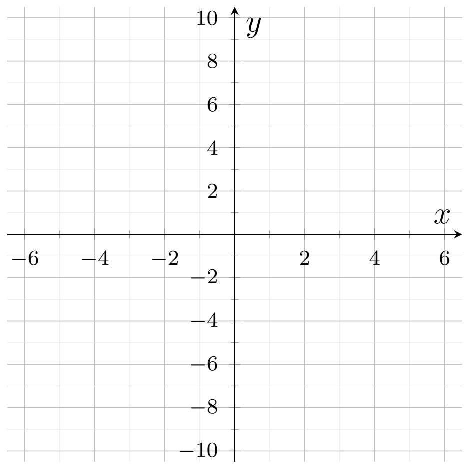
8. $y = \frac{1}{2}(x+2)^2 - 2$. Give the vertex and the two x-intercepts. Will this parabola be narrower or wider than $y = x^2$? Sketch it.



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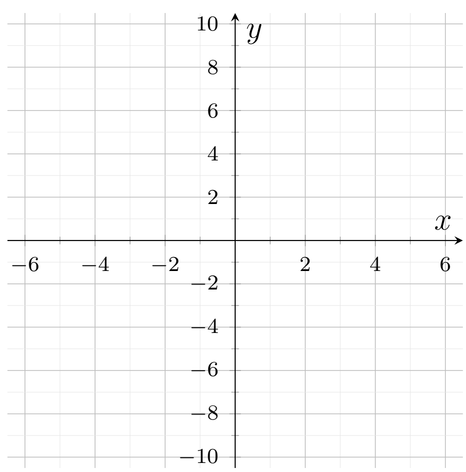
9. $y = x^2 + 2x - 3$. Find the vertex from $x = -\frac{b}{2a}$ and the x-intercepts by factoring, then sketch.



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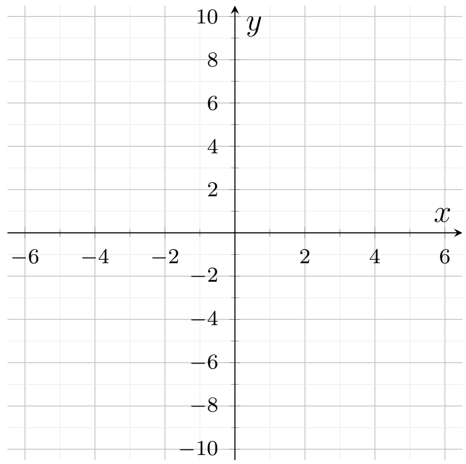
10. $y = 2x^2 - 8x + 6$. Factor out the leading coefficient first, give the x-intercepts, then use the fact that the axis of symmetry sits midway between them to get the vertex. Sketch.



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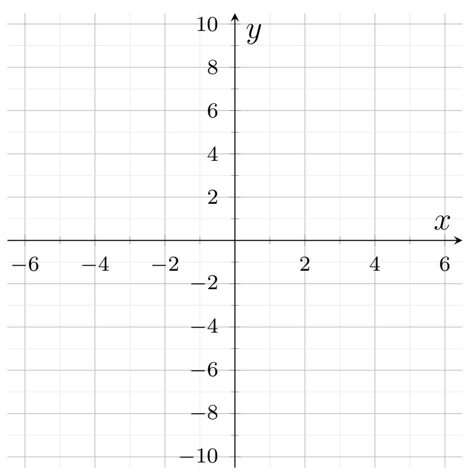
11. $y = x^2 - 6x + 4$. Complete the square to rewrite this in vertex form $y = (x - h)^2 + k$, state the vertex, and sketch.



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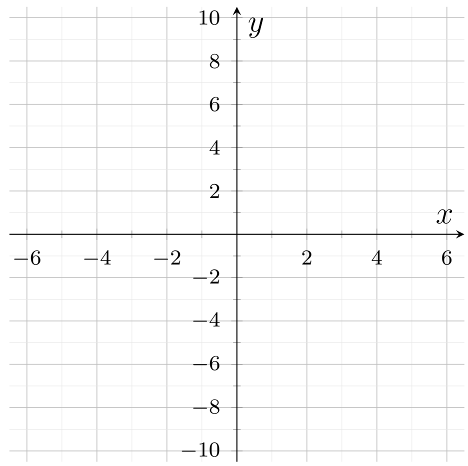
12. A parabola has its vertex at $(3, -4)$ and passes through the point $(5, 0)$. Write its equation in vertex form, checking that $(5, 0)$ really does satisfy your equation. Then sketch it, labelling the vertex, the axis of symmetry and both x-intercepts, and write one sentence explaining how the point $(5, 0)$ pinned down the value of a.



Answer: ____________________